AWS – Network Security

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List of Security options in AWS

- Security Groups
- Network Access Control List (NACL)
- Keys
- Encryptions

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Security Group and NACL

- **These are used to Control the Traffic , IN and OUT of the Sytem.**
- **The Traffic is controlled based on “Port” numbers and “IP address”.**
- **Port number ranges from 1 to 65536.**
- **IP address** can be controlled either as **single IP** or **range of IP’s**

- **SG is a STATEFULL firewall**
- **NACL is a STATELESS firewall**

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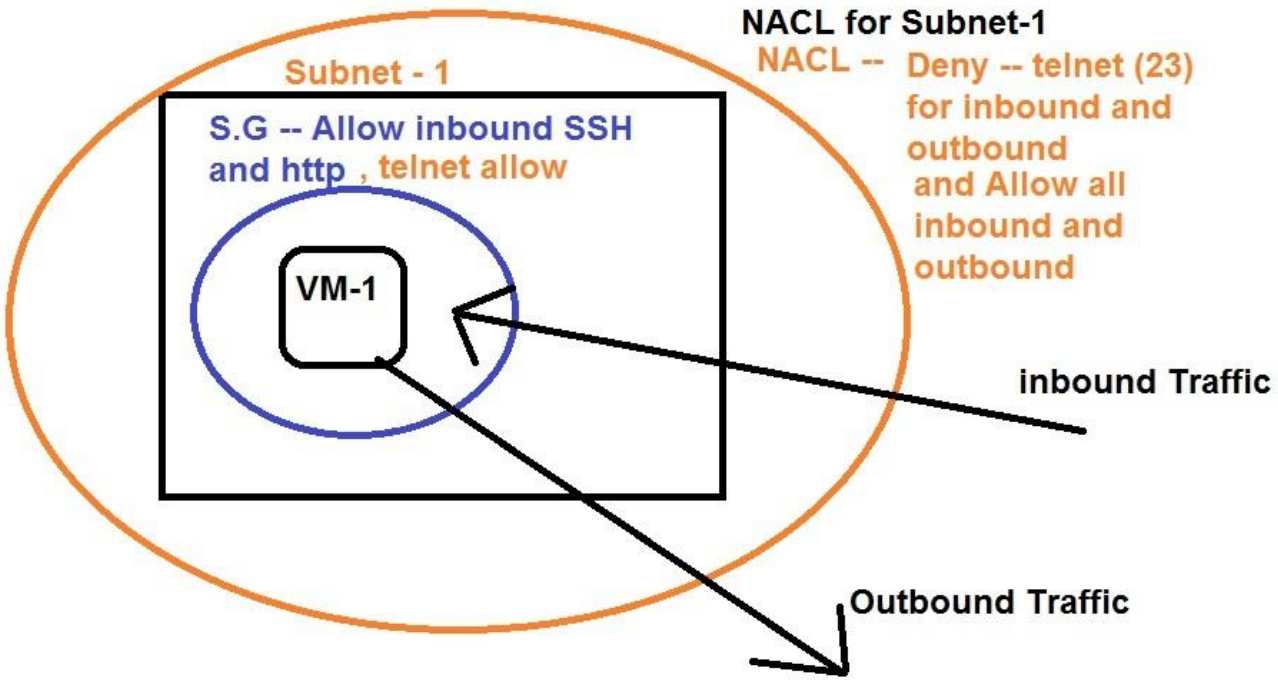

NACL for Subnet-2

Inbound Rules

Allow -- SSH -- from 100.100.100.10
Deny -- SSH -- ALL
Deny -- 8080 -- ALL
Deny -- 9000-9999 -- ALL
Allow -- ALL

Outbound

Deny -- ssh -- ALL
Allow -- ALL

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Security Group(SG) - Properties

- Security Group – We can apply the rule INBOUND or OUTBOUND
- It is applied to a EC2, Load Balancer, EFS, Autoscaling.
- We can apply multiple Security Group to an **Virtual Machine**.
- SG is local to a VPC.
- By default a SG is created for every VPC.
- Default, ALL outbound traffic is allowed in a new Security Group
- Default, All Inbound traffic is denied in a new Security Group.

- **Single Security Group** can be assigned to **Multiple Virtual machine**.
- **Single Virtual Machine** can have **Multiple Security Groups**
- We can Control **both IPV4** and **IPV6** traffic

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SG – Configuration rules

- **Inbound rules are created only with “Source IP” and Source port”**
- **Outbound rules are created only with “Dst IP” and “Dst port”**
- **If you create a rule , that Rule will be allowed.**
- **Source or Destination has 3 option**
  - **Custom – specific Network or IP’s**
  - **My IP address – My Source Public IP (Of my Laptop Internet IP)**
  - **Any IP – All network. (0.0.0.0/0 for IPv4 and ::/0 for IPv6)**

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Security Group

- **Limitation**
- We cannot deny a PARTICULAR traffic in Security Group.

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Example of Security Group

sg-06a7193bc9d359aba | SSH-HTTP-ICMP-8080

Summary | **Inbound Rules** | Outbound Rules | Tags

Edit

Type	Protocol	Port Range	Source	Description
HTTP (80)	TCP (6)	80	0.0.0.0/0	ALLow all for HTTP
HTTP* (8080)	TCP (6)	8080	0.0.0.0/0	
SSH (22)	TCP (6)	22	157.49.174.50/32	allowed from my lapt...
SSH (22)	TCP (6)	22	2409:4071:2318:3ba5:68d1:b28:9f7d:5c11/128	allowed from my lapt...
All ICMP - IPv6	IPv6-ICMP (58)	ALL	::/0	Allow all for ICMP o...
MySQL/Aurora (3306)	TCP (6)	3306	192.168.0.0/16	
All ICMP - IPv4	ICMP (1)	ALL	0.0.0.0/0	Allow all for ICMP

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Network Access Control List - NACL

- **NACL is applied to an Subnet.**
- Default NACL is created when a VPC is created.
- Default NACL has **Allow ALL traffic for both “Inbound” and “OutBound”.**
- NACL is local to a VPC
- We can have separate NACL for individual Subnets.

- Unlike SG, We can Allow or Deny a particular Traffic.
- Also we can assign Rule “numbers” for each rule.

- We can Control both **IPV4 and IPV6** traffic

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NACL – Configuration rules

- **Inbound rules are created only with “Source IP” and Source port”**
- **Outbound rules are created only with “Dst IP” and “Dst port”**
- **If you create a rule , that Rule can be ALLOWED or DEINED .**
- **Source or Destination has 3 option**
  - **Custom – specific Network or IP’s**
  - **My IP address – My Source Public IP (Of my Laptop Internet IP)**
  - **Any IP – All network. (0.0.0.0/0 for IPv4 and ::/0 for IPv6)**

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NACL Example

For 1st 5 Subnets,

Allow ALL traffic outbound and Inbound – Deny SSH all traffic except my laptop IP.

acl-08495ae65c971ad82

Summary | Inbound Rules | **Outbound Rules** | Subnet Associations | Tags

Allows outbound traffic. Because network ACLs are stateless, you must create inbound and outbound rules.

Edit

View: All rules

Rule #	Type	Protocol	Port Range	Destination	Allow / Deny
100	ALL Traffic	ALL	ALL	0.0.0.0/0	ALLOW
101	ALL Traffic	ALL	ALL	::/0	ALLOW
*	ALL Traffic	ALL	ALL	::/0	DENY
*	ALL Traffic	ALL	ALL	0.0.0.0/0	DENY

acl-08495ae65c971ad82

Summary | **Inbound Rules** | Outbound Rules | Subnet Associations | Tags

Allows inbound traffic. Because network ACLs are stateless, you must create inbound and outbound rules.

Edit

View: All rules

Rule #	Type	Protocol	Port Range	Source	Allow / Deny
1000	SSH (22)	TCP (6)	22	157.49.174.50/32	ALLOW
1001	SSH (22)	TCP (6)	22	0.0.0.0/0	DENY
1100	ALL Traffic	ALL	ALL	0.0.0.0/0	ALLOW
1101	ALL Traffic	ALL	ALL	::/0	ALLOW
*	ALL Traffic	ALL	ALL	::/0	DENY
*	ALL Traffic	ALL	ALL	0.0.0.0/0	DENY

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Troubleshooting issue and Other Contents on Security

https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/using-network-security.html

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